Picking the right projection is never easy, and depends on the goals of the map you're making. If you're working with traditional tile mapping, then you'll probably stick with Mercator. If you're working on the world scale, it's usually best to use an equal-area projection like Mollweide that doesn't distort the visual area of geographic features. But because D3 has so many different projections available, you should experiment to see which best suits the particular map you're creating.

> **Infoviz term: choropleth map**
>
> As you encounter more mapmaking, you'll hear the term *choropleth map* used to refer to a map that encodes data using the color of a region. You can use the existing geographic features, in this case countries, to display statistical data, such as the GDP of a country, its population, or its most widely used language. You can do this in D3 either by getting geodata where the `properties` field has that information or by linking a table of data to your geodata where they both have the same unique ID values in common.
>
> Keep in mind that choropleth maps, although useful, are subject to what's known as the *areal unit problem*, which is what happens when you draw boundaries or select existing features in such a way that they disproportionately represent your statistics. This is the case with gerrymandering, when political districts are drawn in such a way as to create majorities for one political party or another.

7.1.4 Interactivity

Much of the geospatial data-related code in D3 comes with built-in functionality that you'll typically need when working with geodata. In addition to determining the area like we did to color our features, D3 has other useful functions. Two that are commonly used in mapping are the ability to quickly calculate the center of a geographic area (known as a *centroid*) and its bounding box, like you see in figure 7.7. In the following listing, you can see how to add mouseover events to the paths we created and draw a circle at the center of each geographic area, as well as a bounding box around it.

Listing 7.8 Rendering bounding boxes with geodata

```
d3.selectAll("path.countries")
.on("mouseover", centerBounds)
    .on("mouseout", clearCenterBounds);

    function centerBounds(d,i) {
      var thisBounds = geoPath.bounds(d);
      var thisCenter = geoPath.centroid(d);
      d3.select("svg")
        .append("rect")
        .attr("class", "bbox")
        .attr("x", thisBounds[0][0])
        .attr("y", thisBounds[0][1])
        .attr("width", thisBounds[1][0] - thisBounds[0][0])
```

Functions of geo.path that give results based on the associated projection

Bounding box is the top-left and bottom-right coordinates as an array